

State of Developer Ecosystem 2024

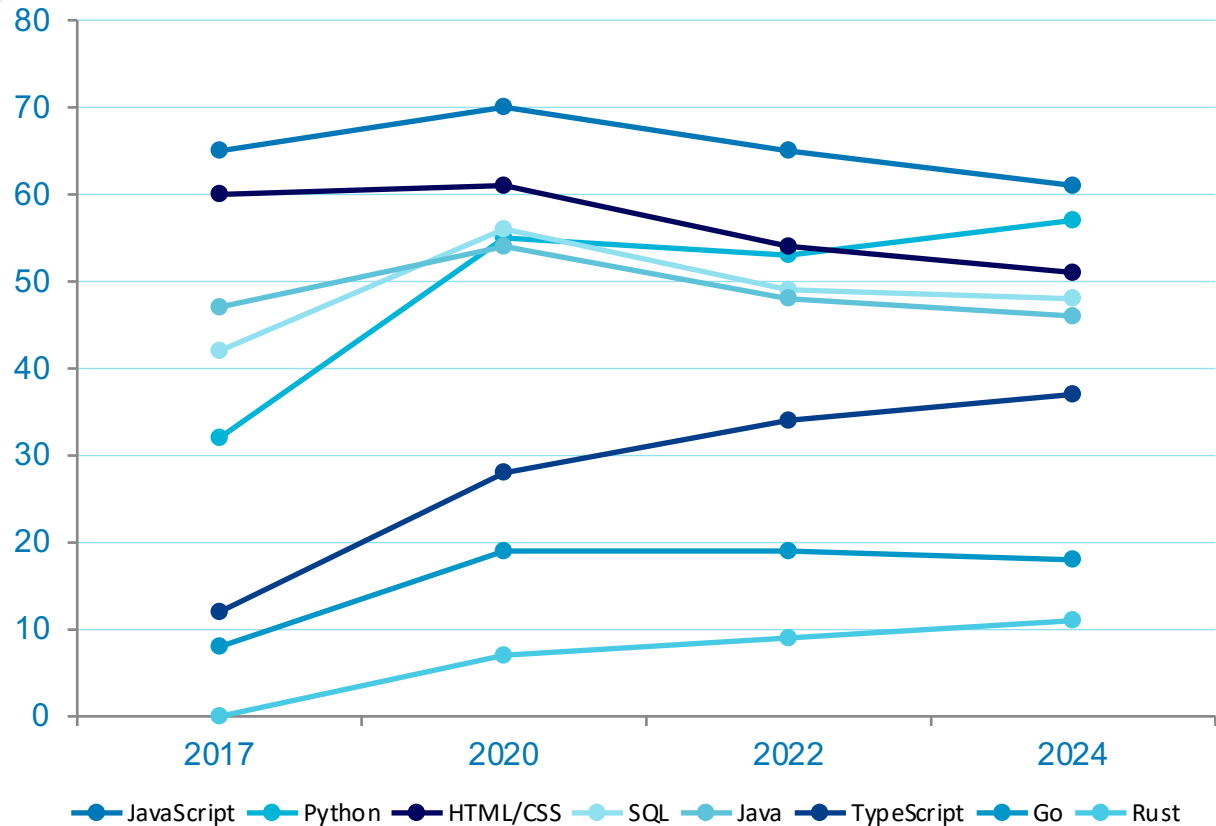
202

Language Trends, Satisfaction, and Strategic Shifts

Based on JetBrains annual survey · 23,262 developers worldwide

4

Python Closes the Gap: 8 Years of Language Usage Shifts



+25pp

Python's surge

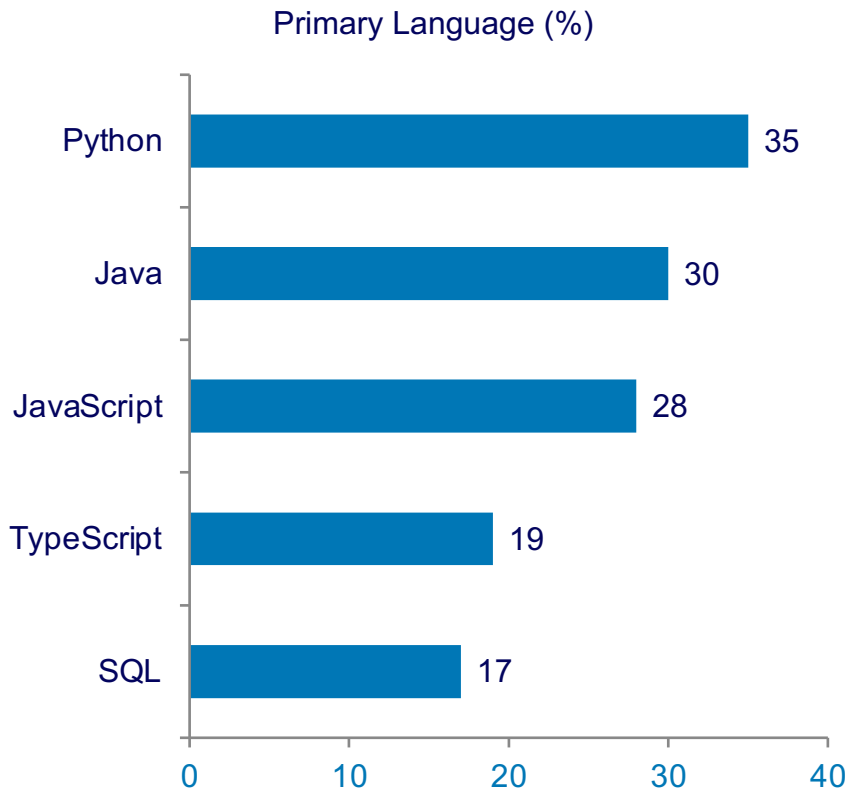
-9pp

JS declining

+25pp

TypeScript x3

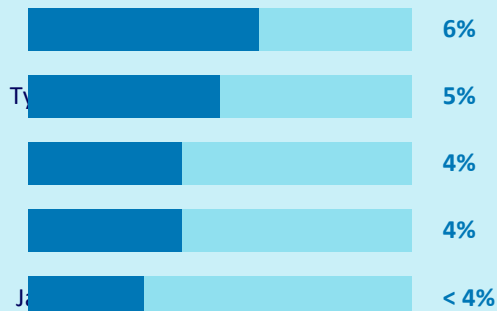
Python Is Now the #1 Primary Language at 35%



35%

Python — most common primary language

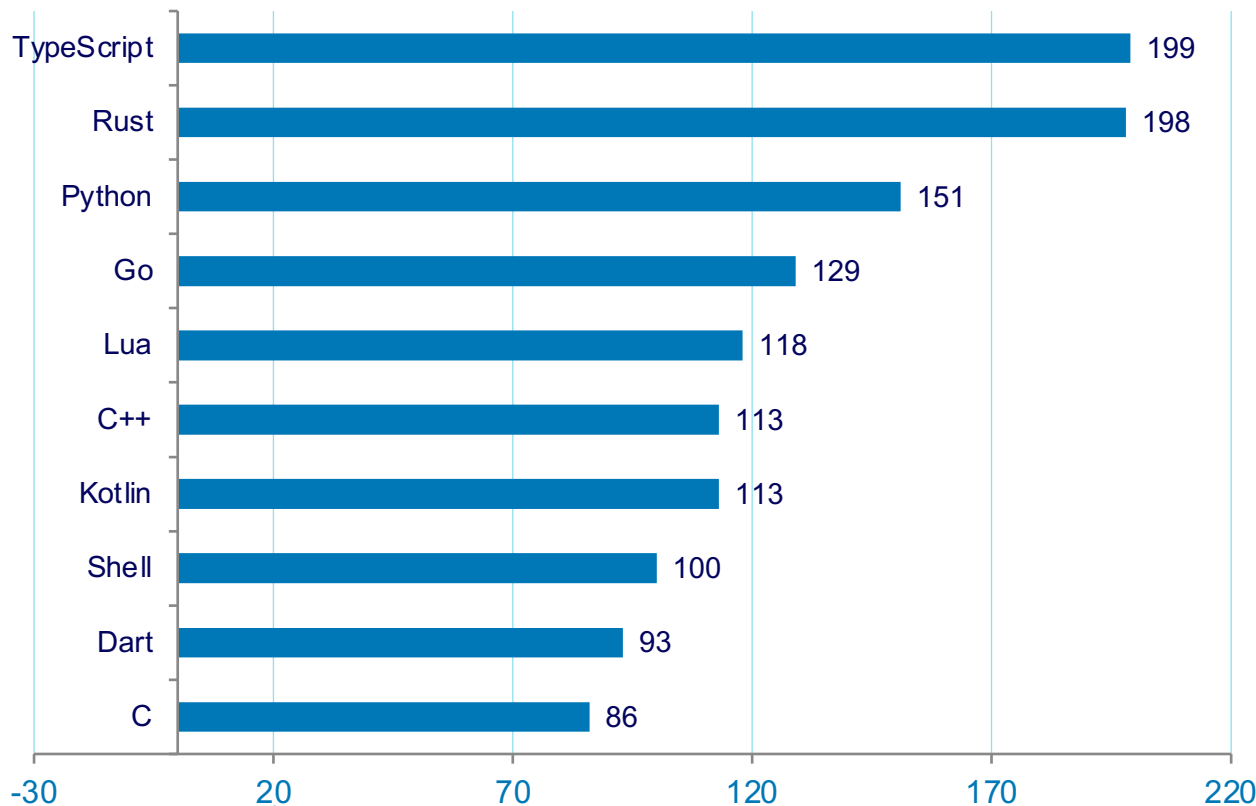
Adoption Intent (% planning to switch)



Developers rarely switch primary stacks — ecosystem lock-in is real

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Satisfaction Score



Source: JetBrains State of Developer Ecosystem 2024

199

TypeScript

Highest satisfaction

Type Safety

TypeScript & Rust — the two most loved languages — both enforce strong type systems

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Application Domain	Top Language	Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%

What This Means for Your Technology Strategy in 2025

1

Invest in Python for AI/ML

Python leads AI/ML (31%), data science (28%), and data processing — the undisputed choice for AI-era teams.

2

TypeScript Owns the Frontend

62% UI share and satisfaction score of 199 — TypeScript is the future of frontend and full-stack.

3

Bet on Rust & Go for Infrastructure

Both lead adoption intent; designed for performance, concurrency, and cloud-native systems.

4

PHP & Ruby Talent Pool is Shrinking

PHP fell from 20% → 17% (2017–2024). Plan migration paths and adjust hiring pipelines.

5

Type Safety = Developer Retention

TypeScript (199) and Rust (198) top satisfaction charts. Type safety correlates with developer happiness — factor into your tooling and hiring strategy.